

RISHABH AGARWAL

Jodhpur
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EDUCATION

Vellore Institute of Technology, Vellore

Jul 2021 - Present

B-Tech, Electronics and Communication Engineering

- GPA: 8.24

St. Edmunds School, Jaipur

Apr 2020 - May 2021

Senior Secondary (XII)

- GPA: 86%

WORK EXPERIENCE

Tata Institute of Fundamental Research

Aug 2023 - Sep 2023

Intern

Ooty, India

- Executed FPGA board programming utilizing Verilog, achieving efficient hardware design and optimized system performance.
- Designed and implemented multiple timers and counters on FPGA, enabling precise time management and enhanced control within the system.
- Integrated a 16x2 LCD with the FPGA board, facilitating clear and immediate data display and enhancing user interface capabilities.

RRSC-West, ISRO

Sep 2023 - Oct 2023

Intern

Jodhpur, India

- Engineered a cutting-edge live temperature monitoring system that delivers real-time data directly to an LCD, ensuring immediate and accurate temperature visualization.
- Integrated an LCD screen with a temperature sensor to display temperature readings in various formats, ensuring versatile and user-friendly data presentation.

PERSONAL PROJECTS

Personal Projects

Illegal Bowling Action Detection

Jul 2024 - Nov 2024

- Developed a real-time web application using Flask to analyze illegal bowling actions, incorporating MediaPipe for precise pose detection and OpenCV for continuous video streaming from the webcam.
- Implemented angle calculations between keypoints (shoulder, elbow, wrist) to determine if the bowler exceeded a 15-degree threshold, providing immediate feedback on potential illegal actions.

Personal Projects

Smart ATS Tracking System

Nov 2023 - Dec 2023

- Designed and developed a Smart ATS application leveraging Streamlit and Google Generative AI to evaluate resumes against job descriptions.
- Implemented PDF parsing functionality using PyPDF2 to extract and process resume content for accurate analysis.
- Crafted a dynamic prompt template to generate precise insights, including JD match percentage, missing keywords, and profile summaries, enhancing resume optimization.

Personal Projects

Smart FIR Portal

Aug 2023 - Sep 2023

- Developed a secure FIR registration system with admin privileges for officers, integrating JWT authentication (1-hour expiry) to ensure 99.9% login success rate and robust access control.
- Integrated browser-based Speech-to-Text APIs in multiple languages, boosting accessibility by 40% and simplifying user grievance submission.
- Used the Gemini APIs for IPC section prediction, achieving a 30% improvement in the accuracy of initial complaint classification.

SKILLS

- Languages: C++, Embedded C, JavaScript, SQL, HTML, CSS
- Frontend & Other skills: React, Tailwind, GenAI, Database Management System, OOPS
- Backend: Node.js, Express.js, MongoDB, PyTorch and TensorFlow

CERTIFICATES

- **Introduction to Internet of Things (05/2022 - 06/2022):** Issued by VIT, Vellore
- **Web Development bootcamp (07/2023 - 08/2023):** Issued by Devtown
- **Google AI Essentials (06/2024 - 07/2024):** Authorized by Google and offered through Coursera

LANGUAGES

- Hindi (Native or Bilingual Proficiency), English (Full Professional Proficiency), Spanish (Limited Working Proficiency)